

AWS

CloudFront & Global Accelerator

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About CloudFront:

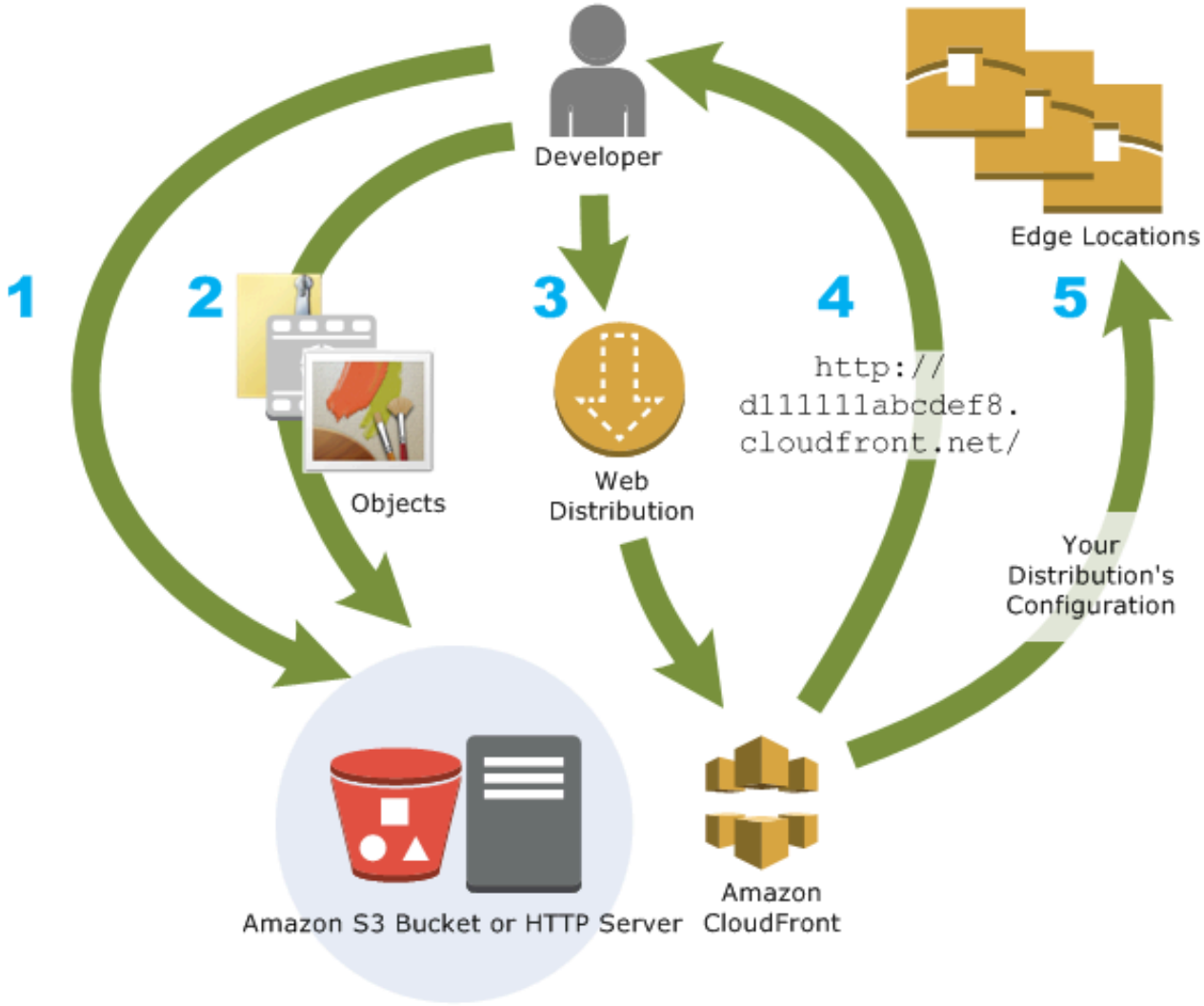
Amazon CloudFront is a web service that speeds up distribution of your static and dynamic web content, such as .html, .css, .js, and image files, to your users. CloudFront delivers your content through a worldwide network of data centers called edge locations. When a user requests content that you're serving with CloudFront, the request is routed to the edge location that provides the lowest latency (time delay), so that content is delivered with the best possible performance.

- If the content is already in the edge location with the lowest latency, CloudFront delivers it immediately.
- If the content is not in that edge location, CloudFront retrieves it from an origin that you've defined—such as an Amazon S3 bucket, a MediaPackage channel, or an HTTP server (for example, a web server) that you have identified as the source for the definitive version of your content.

As an example, suppose that you're serving an image from a traditional web server, not from CloudFront. For example, you might serve an image, sunsetphoto.png, using the URL https://example.com/sunsetphoto.png.

About CloudFront:

Setup CloudFront to deliver content illustrations.



You create a CloudFront distribution to tell CloudFront where you want content to be delivered from, and the details about how to track and manage content delivery. Then CloudFront uses computers—edge servers—that are close to your viewers to deliver that content quickly when someone wants to see it or use it.

Features of CloudFront:

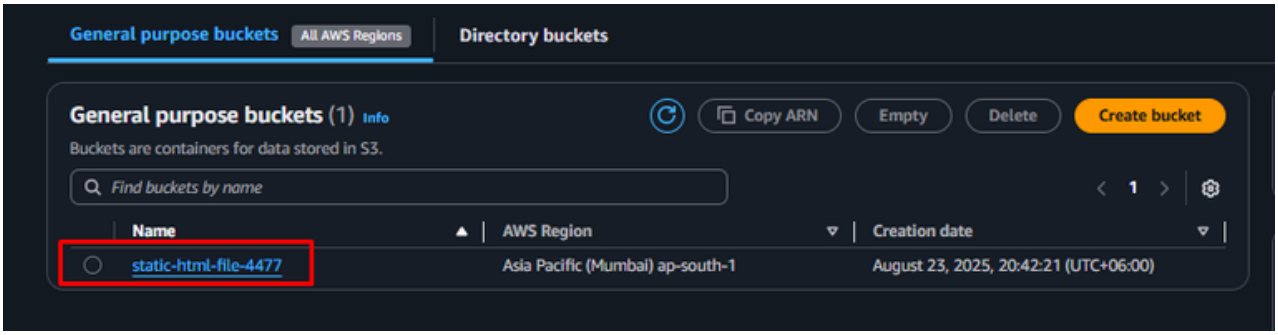
- **Global Edge Network:** 225+ edge locations deliver content closer to users.
- **High Performance:** Low latency and fast data transfer from the nearest edge.
- **Scalable Security:** Built-in AWS Shield (DDoS) & WAF integration.
- **Secure Connections:** HTTPS support & Field-Level Encryption.
- **Content Caching:** Caches static and dynamic content for speed.
- **Flexible Origins:** Supports S3, EC2, or any HTTP backend.
- **Edge Compute:** Lambda@Edge & CloudFront Functions for custom logic.
- **Cost-Effective:** Pay-as-you-go with AWS consolidated billing.
- **Geo-Restrictions:** Control access by user location.

Use case of CloudFront:

- **Website Acceleration:** Speeds up delivery of static & dynamic web content.
- **Video Streaming:** Low-latency live & on-demand video streaming worldwide.
- **API Acceleration:** Delivers APIs and microservices faster and securely.
- **E-commerce Optimization:** Ensures fast, secure, scalable shopping experiences.
- **Global Application Delivery:** Provides consistent performance across regions.
- **DDoS Protection:** Defends applications with AWS Shield integration.
- **Edge Personalization:** Run custom logic (A/B testing, headers, redirects) with Lambda@Edge.
- **Software & Game Distribution:** Reliable, fast downloads of large files.
- **Access Control:** Restrict content using geo-blocking or signed URLs.

Configure CloudFront:

Before configuring CloudFront, let’s create an S3 bucket and host some files to see the results more clearly. We will apply CloudFront to our S3 bucket. Don’t forget to enable static website hosting and set the document path in your bucket.Here the bucket:

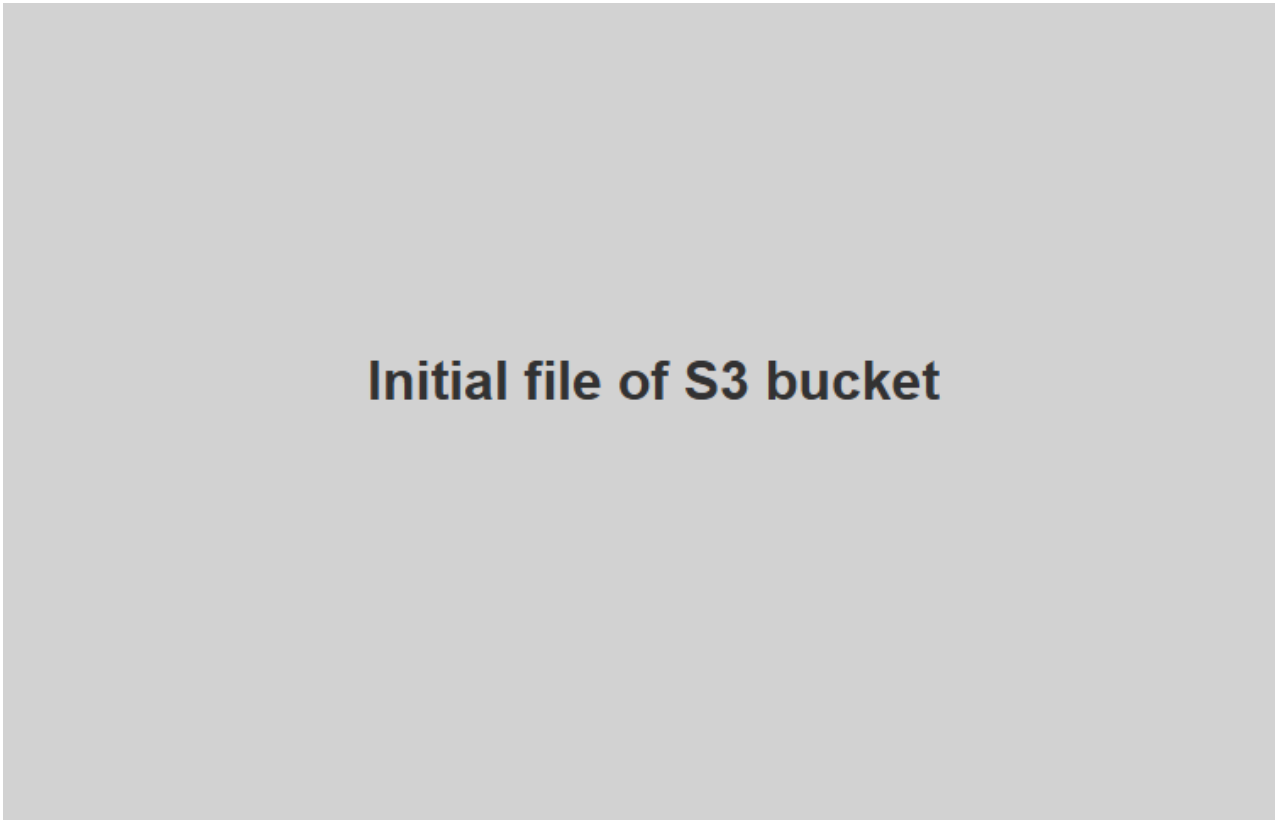


Here the html file code:

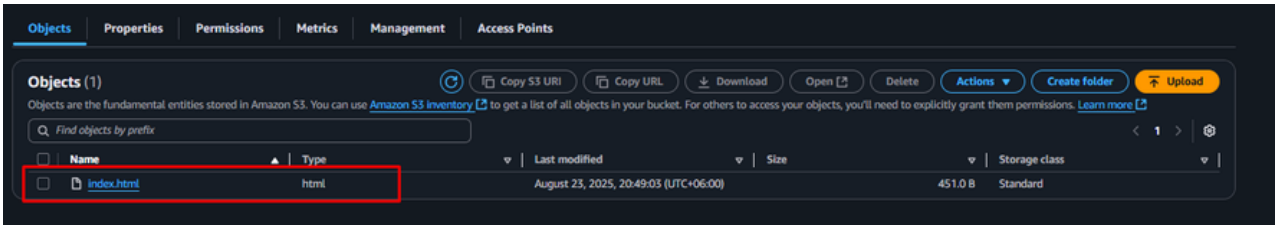
```
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <title>Amazon s3 bucket</title>
 <style>
  body {
   background-color: #d3d3d3;
   display: flex;
   justify-content: center;
   align-items: center;
   height: 100vh;
   margin: 0;
   font-family: Arial, sans-serif;
  }
  h1 {
   color: #333;
  }
 </style>
</head>
<body>
 <h1>Initial file of S3 bucket</h1>
</body>
</html>
```

Configure CloudFront:

HTML code output look like this:

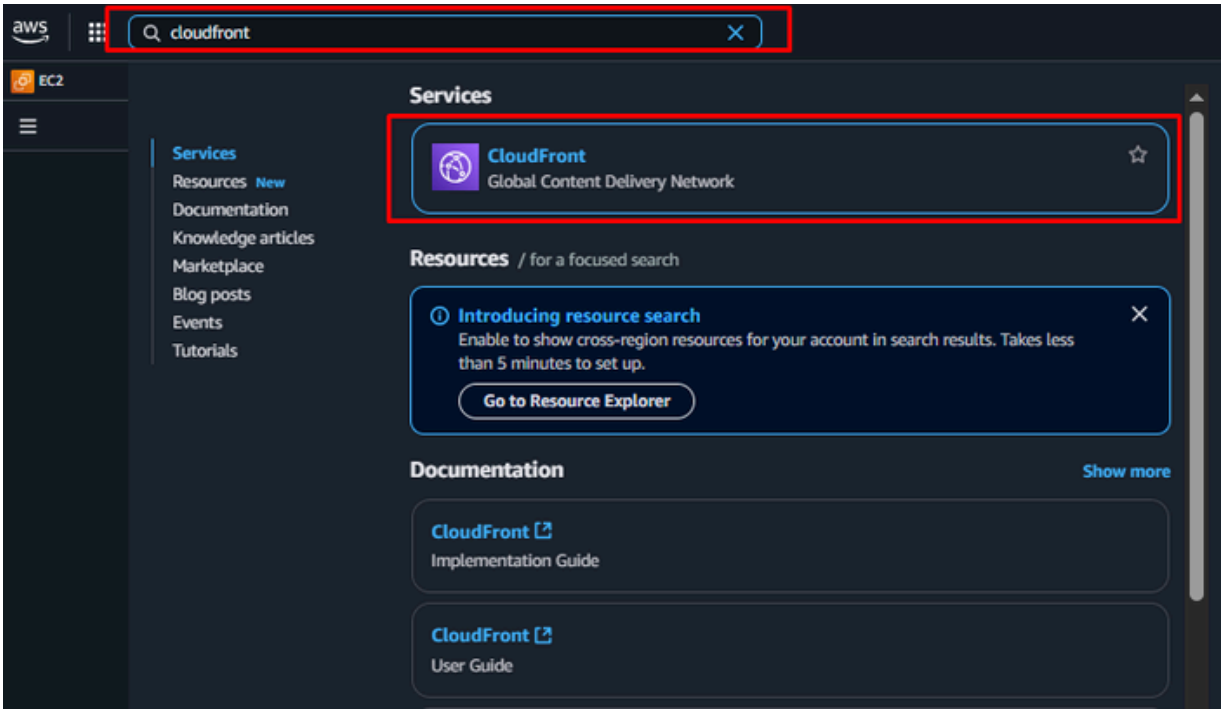


Here the uploaded file:

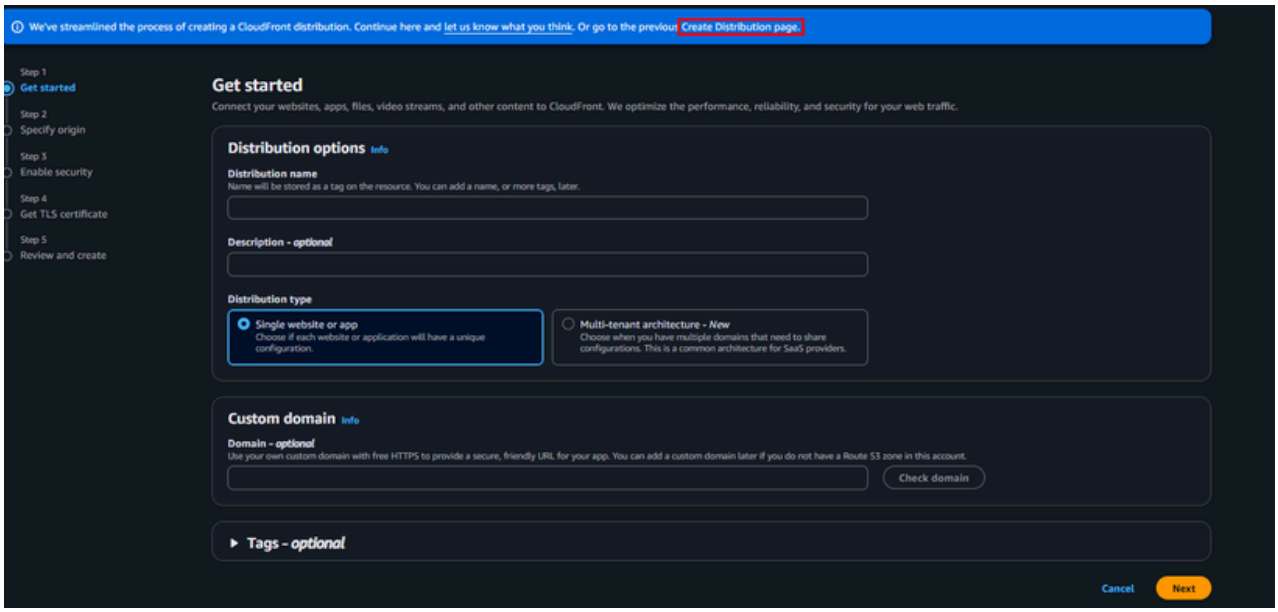


Configure CloudFront:

Now, configure cloudfront with this s3 bucket. For creating cloudfront search and click on cloudfront.

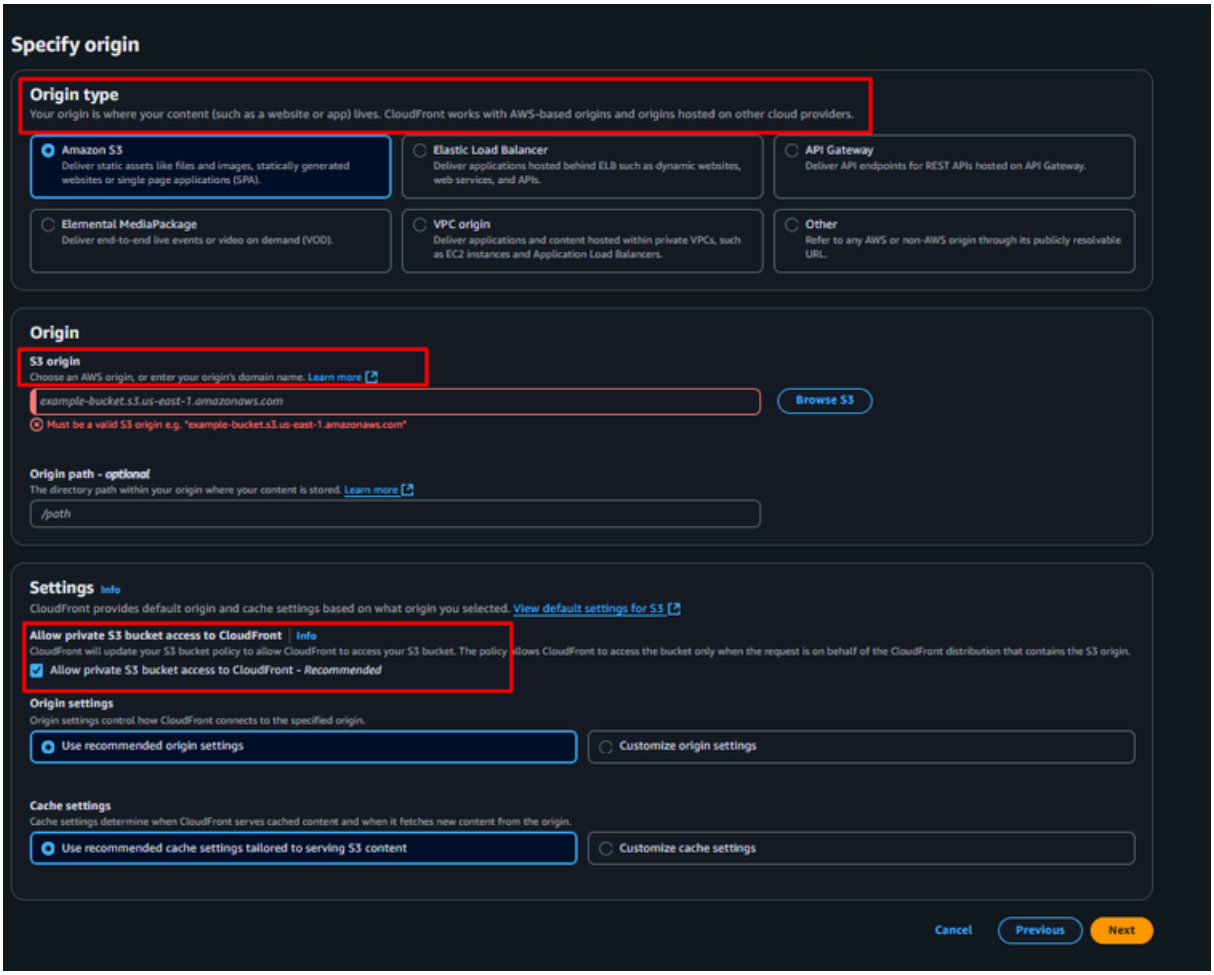


You can create it using current distribution or previous distribution too.

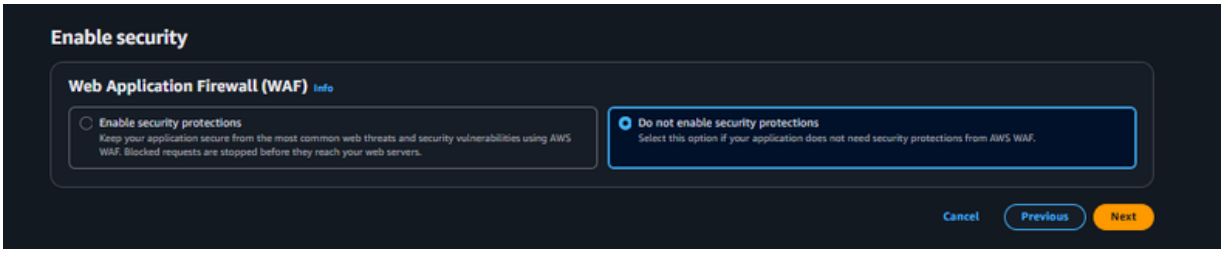


Configure CloudFront:

Choose the origin type, select S3 origin, and then allow CloudFront to access your S3 bucket.

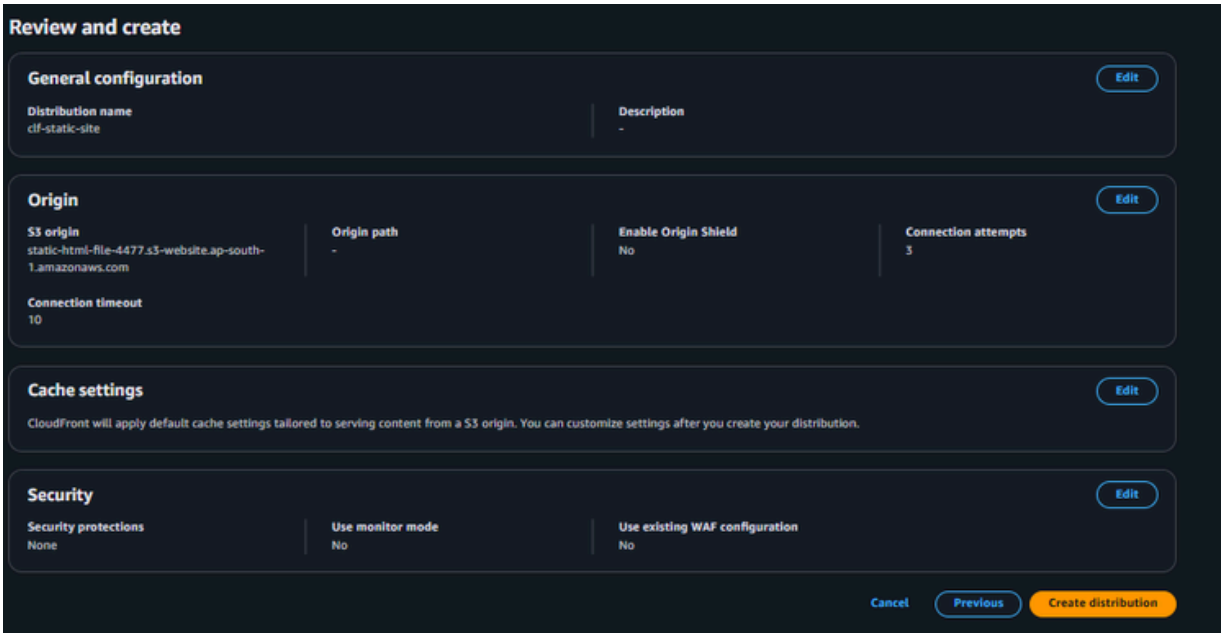


If you need you can enable security. In my case i skip this.

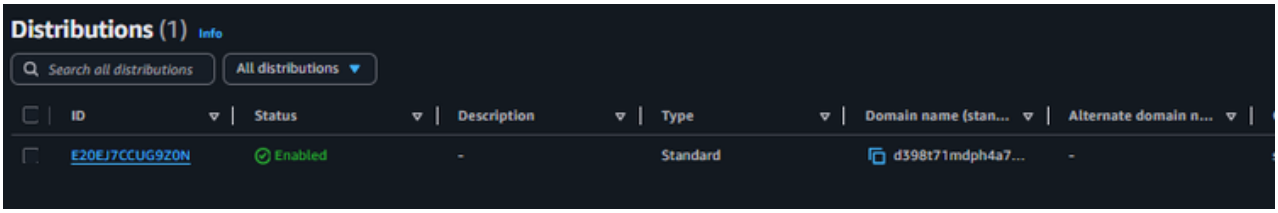


Configure CloudFront:

Now review your setup and create.

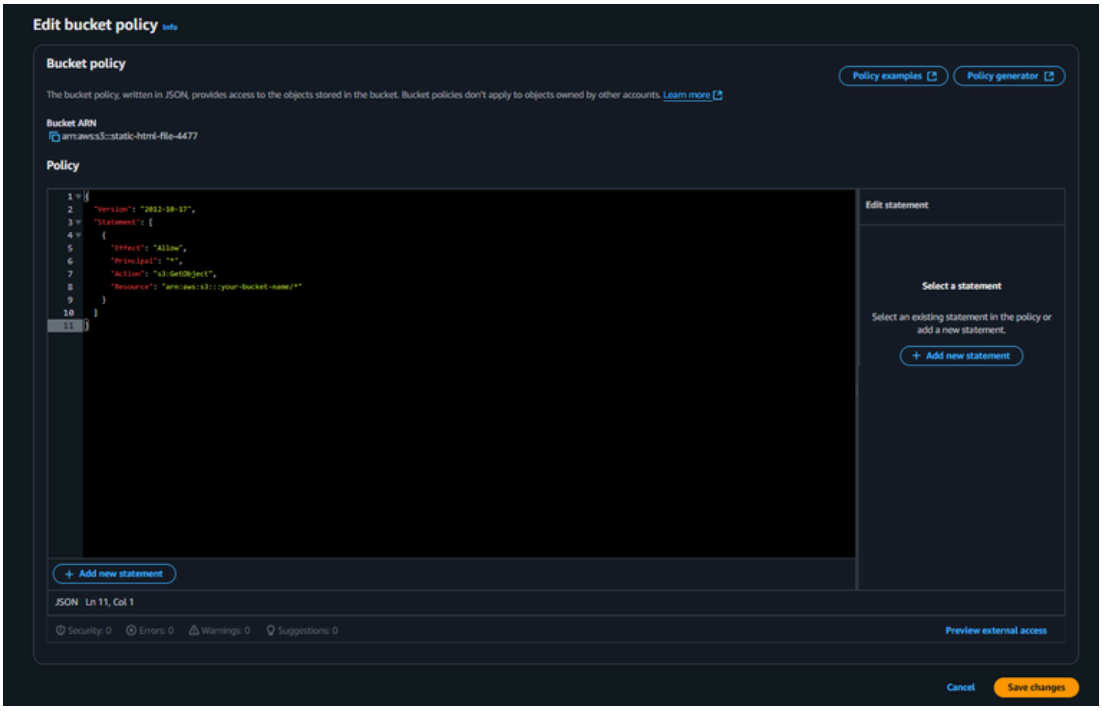


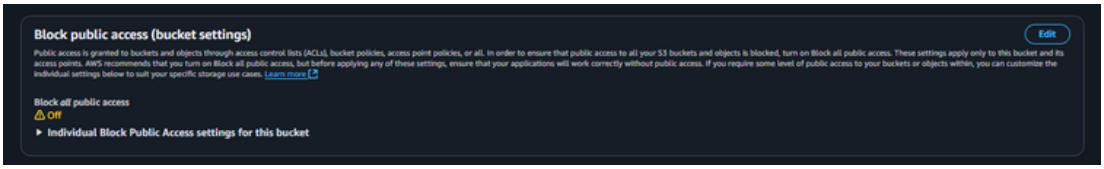
Here is the CloudFront Distribution.



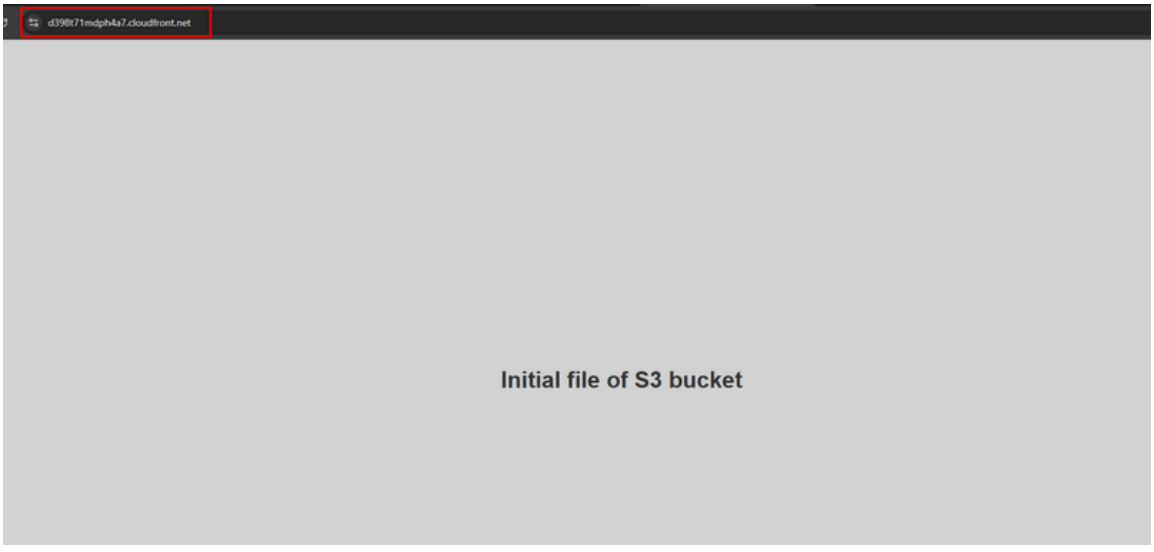
Configure CloudFront:

Now, set the bucket policy for your S3 bucket and also disable block public access. Then, browse your site via CloudFront.





Here the result:

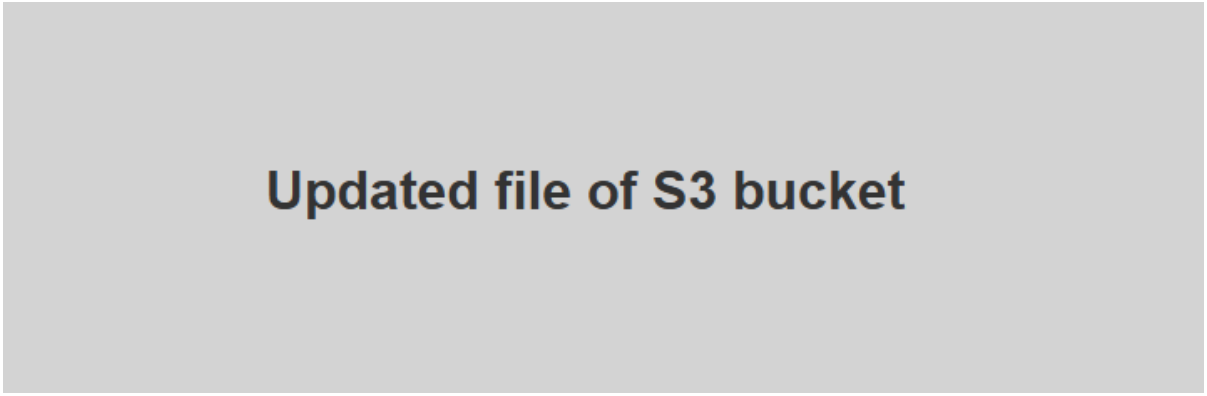


Configure CloudFront:

Now change the s3 bucket file and see the result cloudfront works or not.
updated html file code:

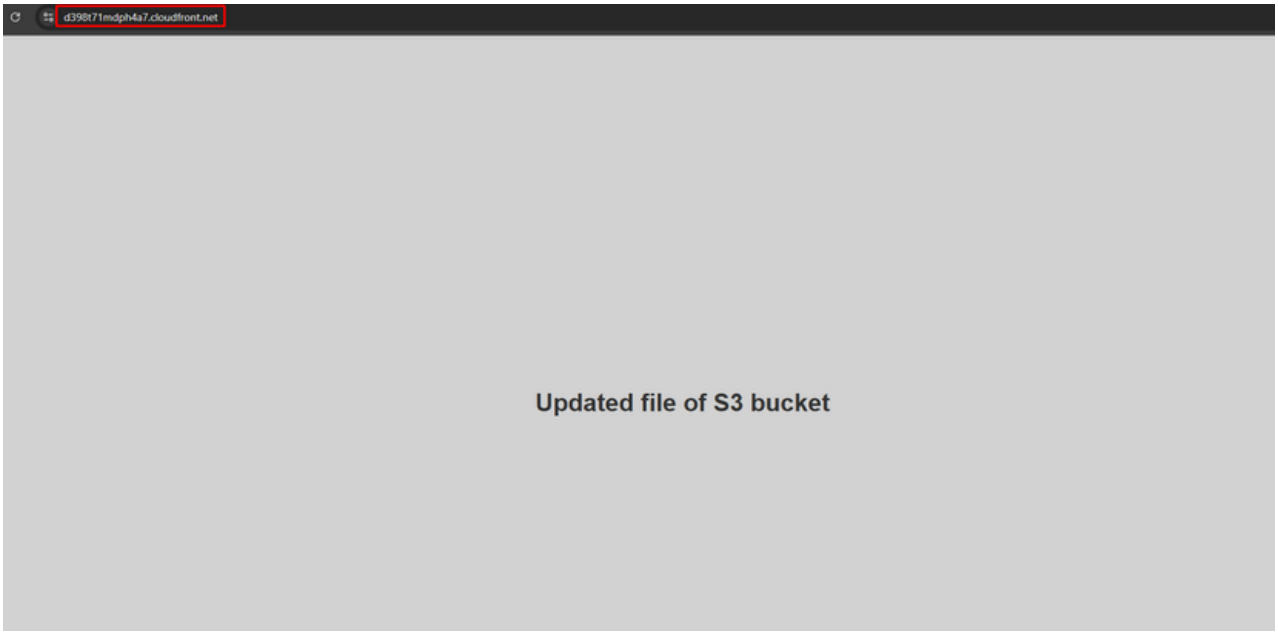
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<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <title>Amazon s3 bucket</title>
 <style>
  body {
   background-color: #d3d3d3;
   display: flex;
   justify-content: center;
   align-items: center;
   height: 100vh;
   margin: 0;
   font-family: Arial, sans-serif;
  }
  h1 {
   color: #333;
  }
 </style>
</head>
<body>
 <h1>Updated file of S3 bucket</h1>
</body>
</html>
```

Here the result of this code:



Configure CloudFront:

Now, you can browse the site via CloudFront. Changes may not appear instantly if you are using an existing distribution. After updating your S3 bucket, you need to clear the CloudFront cache.



CloudFront Cache Invalidations:

Cache invalidation in CloudFront is the process of removing objects from edge caches before they expire, forcing CloudFront to fetch the latest version from the origin.

Features of Cache Invalidations:

Default Expiry:
Objects remain cached until TTL (Time-to-Live) ends.

Invalidation Request:
Let's you manually remove objects before TTL.

Granular Control:
- Invalidate specific files (such as /index.html).
- Use wildcards (such as /* to remove all cached objects).

Propagation:
Changes take a few minutes to update across all edge locations.

Costs:
The first 1,000 invalidation paths/month are free, extra paths incur cost.

Alternatives:
Instead of invalidations, you can:
- Use versioned file names (e.g., app.v2.js) → Best practice.
- Set appropriate Cache-Control / TTL headers.=

About Global Accelerator:

AWS Global Accelerator is a service in which you create accelerators to improve the performance of your applications for local and global users.

Depending on the type of accelerator you choose, you can gain additional benefits:

- With a standard accelerator, you can improve availability of your internet applications that are used by a global audience. With a standard accelerator, Global Accelerator directs traffic over the AWS global network to endpoints in the nearest Region to the client.
- With a custom routing accelerator, you can map one or more users to a specific destination among many destinations.

Features of Global Accelerator:

- **Global Edge Network:** Routes traffic via the nearest AWS edge location for better performance.
- **Static IP Addresses:** Provides two fixed IPs for easy endpoint management.
- **Performance Optimization:** Reduces latency and jitter by using AWS's optimized network paths.
- **Fault Tolerance:** Automatically reroutes traffic to healthy endpoints.
- **TCP Termination:** Ends TCP connections at the edge to lower latency.
- **Continuous Availability Monitoring:** Performs health checks and redirects traffic if needed.
- **Traffic Dials:** Control traffic distribution across regions or endpoints.
- **Client IP Preservation:** Keeps client IPs for stateful applications (custom routing).
- **UDP & TCP Support:** Handles both UDP and TCP traffic for non-HTTP apps.
- **Bring Your Own IP (BYOIP):** Use your own IP ranges for the accelerator entry point.

Use case of Global Accelerator:

- **Global Application Performance:** Improve latency and speed for users worldwide by routing traffic to the nearest AWS edge location.
- **Highly Available Applications:** Automatically reroute traffic to healthy endpoints during failures for continuous availability.
- **Multi-Region Deployments:** Distribute traffic across multiple AWS regions efficiently using traffic dials.
- **Gaming & Media Applications:** Handle both TCP and UDP traffic with low latency for real-time applications.
- **Static IP Requirement:** Provide fixed IP addresses for applications needing consistent endpoints.
- **Stateful Applications:** Preserve client IP addresses for applications that require session awareness.
- **Disaster Recovery:** Enable quick failover to other regions during regional outages.
- **BYOIP (Bring Your Own IP):** Use your existing IP ranges as entry points for enterprise applications.

CloudFront vs Global Accelerator:

AWS CloudFront

- Content Delivery: Caches and delivers static & dynamic content via edge locations.
- HTTP/HTTPS Focused: Optimized for web applications and APIs.
- Latency Reduction: Improves performance mainly through caching and CDN.
- Security Integration: Works with AWS WAF, Shield, and SSL/TLS for secure content delivery.

AWS Global Accelerator

- Global Traffic Management: Routes user traffic to optimal endpoints across regions.
- Static IPs: Provide fixed anycast IPs for easier application access.
- Low Latency & High Availability: Reduces latency and reroutes traffic automatically on failures.
- Protocol Flexibility: Supports TCP and UDP for non-HTTP applications like gaming or media.

Thank You

Stay Connect:

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